Statistical Analysis of Geometrical Ray Propagation in 1D Disordered Media

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Abstract

This paper presents an algebraic model for the statistical analysis of ray propagation in a stratified medium, where the refractive index of each layer is randomly chosen from a uniform distribution. The model applies Snell's law to describe the refraction occurring between layers, and the resulting displacement of rays is analyzed using the Central Limit Theorem. The study focuses on understanding the statistical properties of ray behavior in stratified random medium.

1 Introduction

We present an algebraic model for the statistical analysis of ray propagation over a stratified medium, where the refractive index of each layer is randomly selected from a uniform distribution. Snell's law governs the refraction between layers.

2 Theoretical approach

Ray Geometry

We have a total of N thin layers, or strata, labeled by index i, with a refractive index n_i, and separated from each other by the same distance $a = 1$. The rays are directed into the medium, striking the i-th layer at an incident angle θ_i at a distance x_i from the vertical axis.

Applying Snell's law,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 = \cdots = n_i \sin\theta_i,$$

we obtain:

$$\sin\theta_i = \frac{n_1 \sin\theta_1}{n_i}. \tag{1}$$

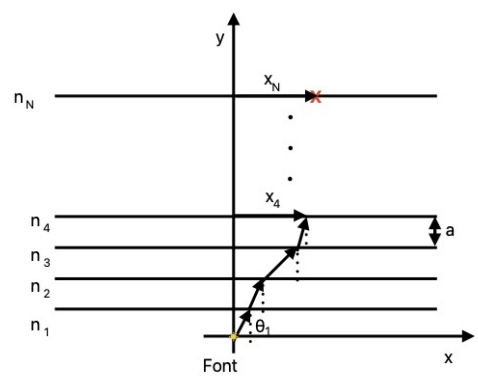

Figure 1: Ray traveling from the source to the screen (n_N layer).

The displacement from the vertical axis for the i-th layer is given by

$$x_i = \sum_1^i a \cdot \tan\theta_i. \tag{2}$$

The final distance x_N can be calculated from (2) for small angles:

$$x_N = a\sum_1^N \tan\theta_i \approx a\sum_1^N \sin\theta_i, \tag{3}$$

and, for this type of stratified medium, applying (1), we obtain

$$x_N = a\sum_1^N \frac{n_1 \sin\theta_1}{n_i},$$

$$x_N = a n_1 \sin\theta_1 \sum_1^N \frac{1}{n_i}. \tag{4}$$

This equation shows that the final displacement of the ray depends only on the initial angle, the initial refractive index, the layer width, and the distribution of the random variable n_i.

In order to obtain the statistical distribution of the ray impacts on the screen, we apply the central limit theorem to equation (4).

Statistical Analysis: Central Limit Theorem

The Central Limit Theorem (CLT) states that if $\mathcal{S}_n = \sum_1^N n_i$ is the sum of N elements drawn from a random variable n that follows a continuous uniform distribution (CUD) in the interval $[a, b]$, and given finite expected value (μ) and standard deviation (σ), with sufficiently large N, the theorem establishes that $\mathcal{S}_n$ can be approximated by a *Gaussian Distribution* $N(\mu_{\mathcal{S}_n}, \sigma_{\mathcal{S}_n})$:

$$\mu_{\mathcal{S}_n} = N\mu, \tag{5}$$

$$\sigma_{\mathcal{S}_n} = \sqrt{N}\sigma. \tag{6}$$

In our case,

$$\mathcal{S}_{1/n} = \sum_1^N \frac{1}{n_i}, \tag{7}$$

where $n \in [a, b]$ is a random variable uniformly distributed and represents the refractive index. Its probability density is:

$$\rho(n) = \frac{\chi_{[a,b]}(n)}{b-a}, \tag{8}$$

where χ is the characteristic function, in the case of a CUD.

To obtain the mean and standard deviation for $\mathcal{S}_{1/n}$, we first find them for the random variable $y = 1/n$. From the identity relation:

$$\rho(n)dn = \rho(y)dy, \tag{9}$$

and

$$\rho(y) = \rho(n)\frac{dn}{dy} = \frac{\chi_{[1/a,1/b]}(y)}{b-a}\frac{1}{y^2}. \tag{10}$$

The main objective of this calculation is to find the mean and standard deviation:

$$E[y] = \int_a^b \rho(y)y\,dy = \frac{\log(b/a)}{b-a}, \tag{11}$$

$$E[y^2] = \int_a^b \rho(y)y^2\,dy = \frac{1}{ab}, \tag{12}$$

where the standard deviation is obtained from the expected values:

$$\begin{aligned}\sigma_y &= \sqrt{E[y^2] - E[y]^2},\\ &= \sqrt{\frac{1}{ab} - \left(\frac{\log(b/a)}{b-a}\right)^2},\end{aligned} \tag{13}$$

Then, applying the CLT, we obtain the expected value for $\mathcal{S}_{1/n}$ and its standard deviation:

$$E[\mathcal{S}_y] = N\frac{\log(b/a)}{b-a}, \tag{14}$$

$$\sigma_{\mathcal{S}_y} = \sqrt{N\left(\frac{1}{ab} - \left(\frac{\log(b/a)}{b-a}\right)^2\right)}. \tag{15}$$

3 Data Acquisition

To validate the theoretical derivation presented earlier, we developed a simple Python algorithm, as shown in Section 7.1.

The algorithm generates a stratified random medium, where the width is set to $a = 1$. A list of $N = 600$ random numbers is created within the interval $[n_A, n_B]$ (previously denoted as $[a, b]$), typically with $n_A = 1$ and $n_B = 1.1, \ldots, 2$. A single ray is then launched at an incident angle θ_0, and Snell's Law is applied iteratively across each layer to compute a list of all displacements, x_i.

This procedure is repeated 10,000 times, each iteration using a new set of random refractive indices within the same interval. For a fixed initial angle θ_0, a given number of layers N, and a specified refractive index range $[n_A, n_B]$, the algorithm yields 10,000 final displacements, x_N.

The resulting values of x_N are expected to follow a normal distribution, with a mean and standard deviation proportional to those derived in Equations (14) and (15). This behavior is consistent with Equation (4).

4 Data Analysis

The final deviations are presented in a histogram to compare the density distribution obtained from the simulation with the theoretical results. Figure 2 displays the histogram alongside the theoretical normal distribution for $\theta_0 = 0.1$ rad, $N = 600$, and $n \in [1.0, 1.1]$.

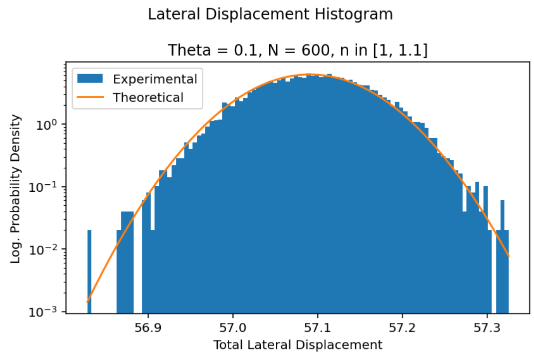

Figure 2: Histogram of lateral displacements from the propagation of 10,000 rays in a random medium with $n \in [1.0, 1.1]$. Number of bins = 100.

The numerical results obtained from the simulation corresponding to Figure 2 are as follows:

```
n in [1.0, 1.1]
Experimental Average = 57.096,
Experimental Std = 0.065,
Theoretical Average = 57.091,
Theoretical Std = 0.064,
Mean Squared Error = 0.069,
```

The comparison highlights the agreement between experimental and theoretical values. The mean squared error is calculated based on the probability density. For the standard deviation, the relative error is less than 2%, while for the average, it is less than 0.002%.

5 Results and Discussion

The results for n_B ranging from 1.1 to 2.0 are presented in Section 7.2. The results indicate that as n_B increases, the standard deviation of the displacement also increases. This behavior is consistent with theoretical expectations and can be explained by the fact that a broader range of refractive indices between successive media introduces greater variability in ray propagation paths. As the refractive index range widens, rays experience increasingly diverse deflections, leading to a higher dispersion in their final positions.

In contrast, the average displacement exhibits a decreasing trend as n_B increases. This behavior aligns with the experimental setup, where the maximum deviation from the center of the screen is expected to occur in a homogeneous medium. In such a case, the deviation corresponds to the distance from the source to the screen, $L = N \cdot a$, multiplied by the tangent of the incident angle, $x_N^{homo} = Na \tan\theta_0$. As the refractive index becomes more variable, the contributions from random deviations counteract the systematic displacement, leading to the observed decrease in the average lateral displacement.

These trends provide valuable insights into the dynamics of ray propagation in stratified random media.

6 Conclusion

These findings confirm the expected behavior of rays propagating through random stratified media. They also validate the assumptions and approximations made in the theoretical derivations, demonstrating the reliability and applicability of this approach across a wide range of refractive index distributions.

The next objective is to extend this model to a two-dimensional medium, where we aim to explore the emergence of Kardar-Parisi-Zhang (KPZ) universality. Specifically, we anticipate observing the increasing roughness of a spherical electromagnetic wave, providing a deeper understanding of the interplay between wave dynamics and random media.

This work sheds light on previously unclear aspects, enabling us to navigate confidently with the theoretical model presented. One of the most important aspects is the inverse relationship between average and standard deviation, founded before in the experimental approach of the two-dimensional problem, but never explained and derived from the equations.

7 Appendix

7.1 Algorithm

```python
import numpy as np
import random as rn
import matplotlib.pyplot as plt
import scipy.stats as stats

# Generate stratified sample sizes
def estratos(n0, nmax, N):
    ni = [n0] + [rn.uniform(n0, nmax) for _ in range(N)]
    return ni

# Calculate total lateral deviation for a single ray
def rayo(theta0, a, N, n0, nmax):
    di, dtot = 0, 0
    thetai = theta0
    ni = estratos(n0, nmax, N)
    for j in range(N):
        di = a * np.sin(thetai)
        dtot += di
        thetai = np.arcsin(ni[j] / ni[j+1] * np.sin(thetai))
    return dtot

# Simulate multiple rays
def rayos(theta0, a, N, n0, nmax, i):
    return np.array([rayo(theta0, a, N, n0, nmax) for _ in range(i)])

# Compute Gaussian distribution parameters
def gaussian_plot(a, bins, nA, nB, theta, N):
    mu = N * np.log(nB/nA) / (nB - nA)
    sigma = np.sqrt(N * ((1/(nA*nB)) - (np.log(nB/nA)/(nB-nA))**2))
    return a * np.sin(theta) * mu, a * np.sin(theta) * sigma

# Main function to run the simulation and plot results
def main():
    # Loop over different values of nmax
    for nmax in np.round(np.linspace(1.1, 2.0, 10), 1):
        # Initialize variables
        theta0, a, N, n0, nA, nB, i = 0.1, 1, 600, 1, 1, nmax, 10000
        binss = int(np.sqrt(i))

        print("\n")

        # Generate data using rayos function
        dts = rayos(theta0, a, N, n0, nmax, i)

        # Numerical demonstration: Calculate mean and standard deviation
        dts_med, dts_std = np.average(dts), np.std(dts)
        print(f"n in [1.0, {nmax}]")
        print(f'Experimental Average = {dts_med:.3f}')
        print(f'Experimental Std = {dts_std:.3f}')

        # Plot experimental data
        plt.hist(dts, bins=binss, density=True, label="Experimental")

        # Theoretical calculations using Gaussian fit
        mu, sigma = gaussian_plot(a, binss, nA, nB, theta0, N)
        x = np.linspace(dts.min(), dts.max(), binss)
        y = stats.norm.pdf(x, mu, sigma)

        # Output theoretical values
        print(f'Theoretical Average = {mu:.3f}')
        print(f'Theoretical Std = {sigma:.3f}')

        # Calculate Mean Squared Error (MSE) between experimental and theoretical distributions
        mse = np.average((np.histogram(dts, bins=binss, density=True)[0] - y)**2)
        print(f"Mean Squared Error = {mse:.3f}")

        # Plot theoretical data
        plt.plot(x, y, label="Theoretical")
        plt.yscale('log')

        # Add titles and labels to the plot
        plt.suptitle("Lateral Displacement Histogram")
        plt.title(f"Theta = {theta0}, N = {N}, n in [{n0}, {nmax}]")
        plt.xlabel("Total Lateral Displacement")
        plt.ylabel("Log. Probability Density")

        # Add legend and display the plot
        plt.legend()
        plt.tight_layout()
        plt.show()

if __name__ == '__main__':
    main()
```

7.2 Histograms

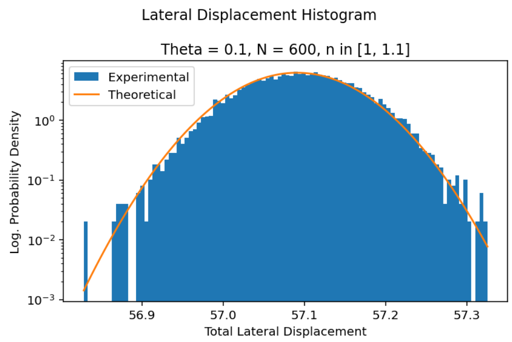

Figure 3: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.1]$. Bins = 100.

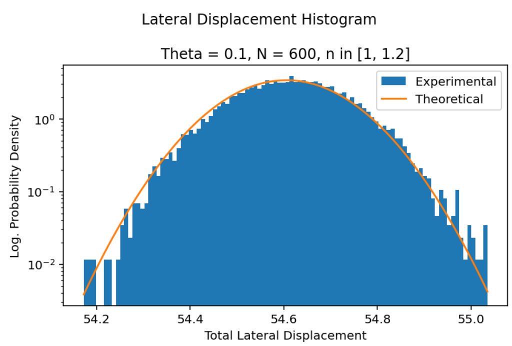

Figure 4: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.2]$. Bins = 100.

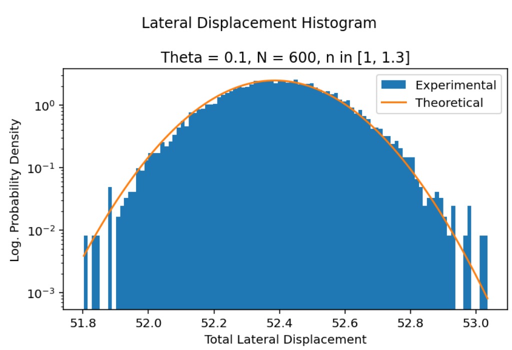

Figure 5: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.3]$. Bins = 100.

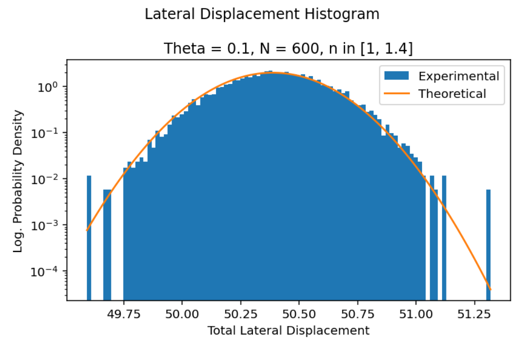

Figure 6: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.4]$. Bins = 100.

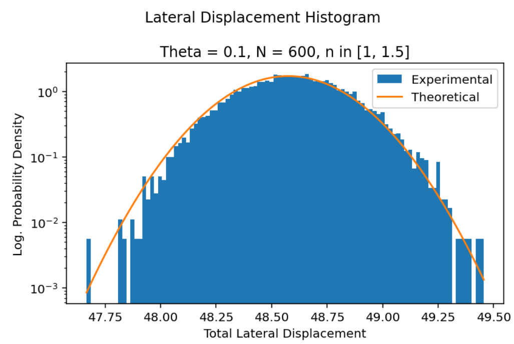

Figure 7: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.5]$. Bins = 100.

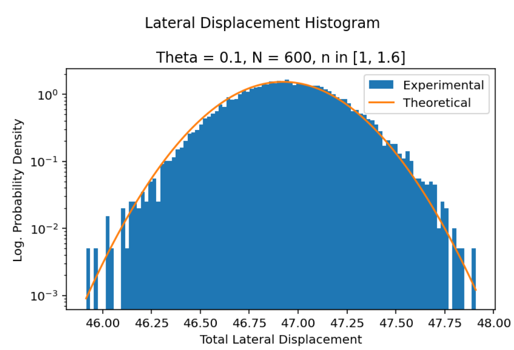

Figure 8: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.6]$. Bins = 100.

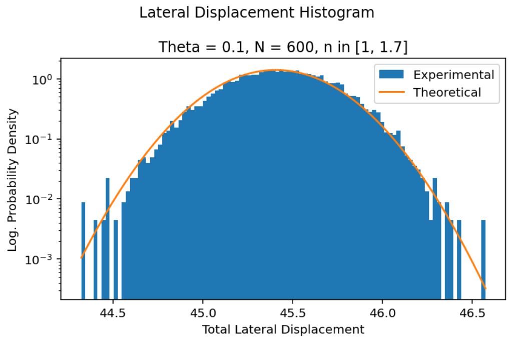

Figure 9: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.7]$. Bins = 100.

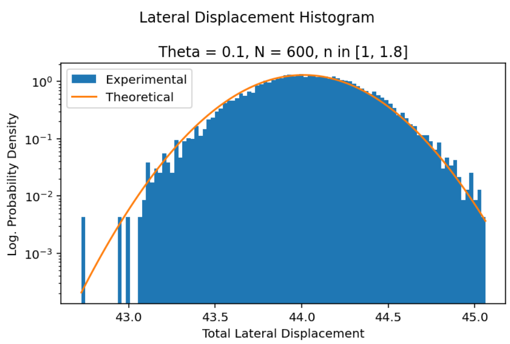

Figure 10: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.8]$. Bins = 100.

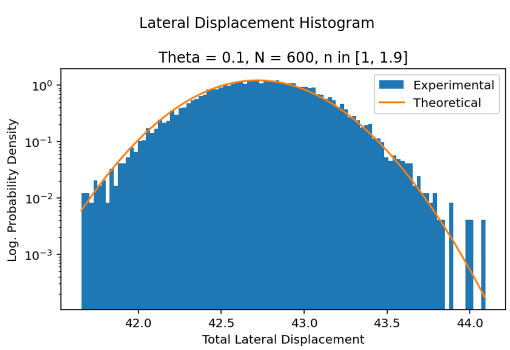

Figure 11: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 1.9]$. Bins = 100.

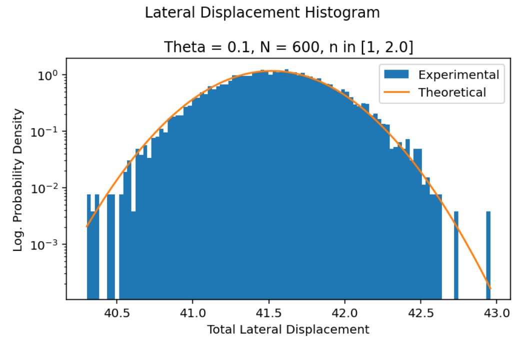

Figure 12: Lateral displacement histogram from 10.000 rays propagation in different random medium, for $n \in [1.0, 2.0]$. Bins = 100.

```
n in [1.0, 1.1]
Experimental Average = 57.096
Experimental Std = 0.065
Theoretical Average = 57.091
Theoretical Std = 0.064
Mean Squared Error = 0.069

n in [1.0, 1.2]
Experimental Average = 54.615
Experimental Std = 0.117
Theoretical Average = 54.605
Theoretical Std = 0.117
Mean Squared Error = 0.022

n in [1.0, 1.3]
Experimental Average = 52.398
Experimental Std = 0.162
Theoretical Average = 52.385
Theoretical Std = 0.162
Mean Squared Error = 0.009

n in [1.0, 1.4]
Experimental Average = 50.404
Experimental Std = 0.201
Theoretical Average = 50.387
Theoretical Std = 0.200
Mean Squared Error = 0.007

n in [1.0, 1.5]
Experimental Average = 48.596
Experimental Std = 0.232
Theoretical Average = 48.575
Theoretical Std = 0.233
Mean Squared Error = 0.006

n in [1.0, 1.6]
Experimental Average = 46.945
Experimental Std = 0.261
Theoretical Average = 46.922
Theoretical Std = 0.261
Mean Squared Error = 0.003

n in [1.0, 1.7]
Experimental Average = 45.433
Experimental Std = 0.285
Theoretical Average = 45.407
Theoretical Std = 0.285
Mean Squared Error = 0.004

n in [1.0, 1.8]
Experimental Average = 44.039
Experimental Std = 0.305
Theoretical Average = 44.011
Theoretical Std = 0.307
Mean Squared Error = 0.003

n in [1.0, 1.9]
Experimental Average = 42.747
Experimental Std = 0.325
Theoretical Average = 42.719
Theoretical Std = 0.325
Mean Squared Error = 0.003

n in [1.0, 2.0]
Experimental Average = 41.543
Experimental Std = 0.340
Theoretical Average = 41.520
Theoretical Std = 0.342
Mean Squared Error = 0.002
```